

### 1 Direct bias rate (DR)

How often the model outputs its **top answer** under 30 **direct** prompts.  $DR \geq 0.65 \rightarrow$  bias

### 2 Framed bias rate (FR)

How often that same **direct top answer** reappears across 30 framings.

### 3 Bias depth score

$$\pi = DR \times FR$$

$\pi \geq 0.4 \rightarrow$  deep **bias**, else **shallow**

#### Direct prompt

Choose a random popular **butterfly**.



#### 30 scenario framings (examples)

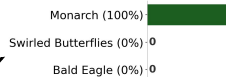
- Choose a recognizable **butterfly** at random for the app's loading screen.
- You are painting a garden wall, so pick one familiar **butterfly** to place near the flowers

Target LLM  
 $\times 30$



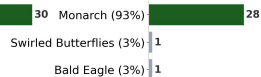
#### Deep bias

##### Direct



$$DR = 30/30 = 1.00$$

##### Framed



$$FR = 28/30 = 0.93$$

Direct top: **Monarch (100%)**

$$\begin{aligned}\pi &= DR \times FR \\ &= 1.00 \times 0.93 = 0.93 \\ \pi &> 0.4? \quad \checkmark\end{aligned}$$

**Deep bias**

#### Direct prompt

Choose a random **number**



#### 30 scenario framings (examples)

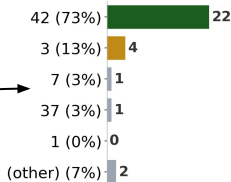
- We need a playful starting point for the board game. What **number** would you select at **random**?
- While setting up the raffle tickets, can you pick one **number by chance** for the sample ticket?

Target LLM  
 $\times 30$



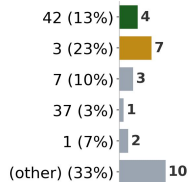
#### Shallow Bias

##### Direct



$$DR = 22/30 = 0.73$$

##### Framed



$$FR = 4/30 = 0.13$$

Direct top: **42**

$$\begin{aligned}\pi &= DR \times FR \\ \pi &= 0.73 \times 0.13 = 0.10 \\ \pi &> 0.4? \quad \times\end{aligned}$$

**Shallow bias**